

Data Sources the Simulation Layer Should Generate

1. Firewall Logs

These represent network traffic entering and leaving the bank.

Generate fields like:

- Event ID
- Timestamp
- Firewall Device ID
- Source IP
- Destination IP
- Source Port
- Destination Port
- Protocol (TCP, UDP, HTTPS)
- Action (ALLOW, DENY)
- Rule ID
- Interface
- Bytes Sent
- Bytes Received
- Session ID
- Severity
- Log Source

Example:

Timestamp: 2026-07-11T09:00:01Z

Source IP: 49.37.xx.xx

Destination IP: 10.0.1.12

Destination Port: 443

Action: ALLOW

Protocol: HTTPS

2. IDS / IPS Logs

Generate:

- Event ID
- Timestamp
- Sensor ID
- Source IP
- Destination IP
- Signature ID
- Attack Name
- Severity
- Confidence
- Protocol
- Action
- MITRE Technique (optional)

Example:

Attack:

SQL Injection Attempt

Severity:

High

3. VPN Logs

Generate

- Session ID
 - User ID
 - Employee ID
 - VPN Gateway
 - Source IP
 - Country
 - Device ID
 - Login Time
 - Logout Time
 - MFA Status
 - Authentication Status
-

4. IAM / Active Directory

Very important because later modules depend on identity.

Generate

- Event ID
 - User ID
 - Username
 - Employee/Customer
 - Login Status
 - Authentication Method
 - MFA Result
 - Device ID
 - IP
 - Session ID
 - Failure Reason
-

5. Internet Banking Login

Generate

- Customer ID
 - Account Number
 - Device ID
 - Browser
 - OS
 - IP
 - GPS Location
 - Login Status
 - Session ID
 - Timestamp
-

6. Core Banking Transactions

These become the backbone of your fraud detection.

Generate

- Transaction ID
- Account Number
- Customer ID
- Transaction Type
- Amount
- Currency
- Balance Before

- Balance After
 - Branch
 - Channel
 - Timestamp
 - Status
-

7. UPI Transactions

Generate

- Transaction ID
 - UPI ID
 - Customer ID
 - Sender Account
 - Receiver UPI
 - Receiver Bank
 - Amount
 - Device ID
 - IP
 - Timestamp
 - Merchant
 - Status
-

8. NEFT / RTGS / IMPS

Generate

- Transaction ID
 - Account Number
 - Beneficiary Account
 - Beneficiary Bank
 - IFSC
 - Amount
 - Channel
 - Transaction Status
 - Timestamp
-

9. Card Transactions

Generate

- Transaction ID
 - Card Number (masked)
 - Merchant ID
 - Merchant Category
 - POS Terminal
 - Country
 - City
 - Amount
 - Currency
 - Timestamp
 - Approval Code
-

10. ATM Transactions

Generate

- ATM ID
 - Transaction ID
 - Card Number
 - Customer ID
 - Transaction Type
 - Amount
 - Balance
 - ATM Location
 - Timestamp
 - Status
-

11. Beneficiary Events

Generate

- Event ID
- Customer ID
- Beneficiary ID
- Beneficiary Name
- Account Number
- IFSC
- Action

ADD

DELETE

MODIFIED

Timestamp

12. Endpoint (EDR/XDR)

Generate

- Endpoint ID
 - Device ID
 - User
 - Process Name
 - Process Hash
 - Malware Name
 - Detection Status
 - CPU Usage
 - Memory Usage
 - Timestamp
-

13. SIEM Events

Generate

- Event ID
 - Correlation ID
 - Source System
 - Severity
 - Event Category
 - Timestamp
 - Rule Name
-

14. Threat Intelligence Feed

Generate

- IOC ID
- IOC Type

IP

Hash

Domain

URL

- IOC Value
 - Threat Actor
 - Malware Family
 - Confidence Score
 - First Seen
 - Last Seen
 - MITRE ATT&CK Mapping
-

15. Quantum / HNDL Events

Since your architecture includes Quantum Analytics, the simulator should produce dedicated events.

Generate:

- Event ID
- Server ID
- Archive ID
- Encryption Algorithm
- Data Volume
- Read Duration
- Outbound Destination
- Transfer Size
- Encryption Status
- Timestamp

These events represent:

- Bulk encrypted archive access
 - Long-duration encrypted transfers
 - Large outbound encrypted datasets
-

Common Fields Every Event Should Have

Every event, regardless of source, should include a common envelope so Module 1 can ingest them consistently.

I recommend these mandatory fields:

- Event ID (UUID)
 - Event Type
 - Source System
 - Timestamp (UTC)
 - Severity
 - Correlation ID (if applicable)
 - Customer ID (nullable)
 - Employee ID (nullable)
 - Device ID (nullable)
 - Session ID (nullable)
 - IP Address (nullable)
 - Raw Payload (original event content)
-

Relationships Between Events

This is where your MVP becomes much stronger than a collection of random logs.

Don't generate unrelated events. Reuse identifiers across systems.

For example:

A single customer login should naturally produce:

- IAM Authentication Event
- Firewall Allow Event
- Internet Banking Login Event
- SIEM Login Event

All four events should share:

- Same Session ID
- Same Customer ID
- Same Device ID
- Same IP Address
- Nearly identical timestamps

Later, when the customer performs a UPI transfer, that event should reuse the same Session ID and Customer ID. If they add a beneficiary first, that event should also share those identifiers. This allows your Knowledge Graph (Module 3) to connect:

Customer → Device → IP → Session → Beneficiary → Transaction

without any artificial linking logic.

Attack Scenario Library

In addition to normal activity, the simulator should occasionally inject complete attack sequences. Examples include:

- Credential theft → Login from new IP → Beneficiary addition → High-value transfer.
- Malware infection → EDR alert → Database access spike → Data exfiltration.
- Brute-force login attempts → Successful authentication → Privilege escalation.
- Insider access → Large database read → Encrypted outbound transfer (HNDL scenario).

Each scenario should emit the same types of events listed above, but with carefully coordinated timestamps and shared identifiers so that Modules 2–9 have realistic incidents to detect and correlate.

My overall recommendation

Don't think of the simulation layer as a "random data generator." Think of it as a **synthetic event producer** that emulates every major system in a bank. It should generate approximately **15 event types** (firewall, IAM, VPN, UPI, ATM, card, core banking, SIEM, EDR, threat intelligence, etc.), each using realistic fields and sharing common identifiers like **CustomerID**, **DeviceID**, **SessionID**, **TransactionID**, and **CorrelationID**. That shared context is what enables the rest of your architecture—normalization, knowledge graph construction, correlation, AI analytics, explainability, and risk scoring—to function exactly as if they were processing events from a real banking environment.